

19 *Use of the Data Booklet is relevant to this question.*

When anhydrous zinc nitrate, $\text{Zn}(\text{NO}_3)_2$, is heated, it decomposes producing three products. Zinc oxide, ZnO , is the residue; nitrogen dioxide, NO_2 , and oxygen are given off.

A 0.020 mol sample of anhydrous zinc nitrate is heated until it fully decomposes.

What is the total volume of gas produced, measured at r.t.p.?

A 0.24 dm³

B 0.96 dm³

C 1.2 dm³

D 2.4 dm³